CHAPTER 7

Wax Denture Fabrication and Try-in to Finishing

1 Buccal polished surface form to achieve suction (provision of moderate concave)

Impression details must be precisely reflected in the resin base through cast stone models. As long as the area about 5.0 mm coronally ranged from the denture base border and the polished surface form on the base is given to a dental technician. But a technician would normally wax up without knowing about patient's individual functional specialty on the posterior buccal polished surface. In fact, similar waxing pattern is normally repeated. Also it is true that no detailed information on the buccal polished surface is given from dentists.

The occlusal force of edentulous jaws, in general, has only 1/6~1/10 the efficiency/power of healthy dentate jaws. In a edentulous patients mouth with a small exertion of force, they can hardly achieve the equivalent function of buccal mucosa as in a healthy dentate mouth. Fig shows comparative photos of different movements done by a healthy dentate mouth and the edentulous mouth. Contouring masses of muscles and magnitude of movements around the lips and cheeks are obviously different (**Fig 7-1**).

In consequence, it can be presumed that the denture buccal polished surface form in most edentulous people will exhibit a different shape from healthy dentate people. That is, if a similar dento-alveolar morphology of healthy dentate jaw is given to the denture buccal surface, food debris will be trapped more easily on the dento-buccal surface because their intraoral activity has decreased so much.

Difference in the intraoral activity between dentulous and edentulous mouth

Highly active dentulous mouth | Edentulous mouth



(1) Purse the lips.



(2) Retract the corners of the mouth.

Fig 7-1

[1] Form alterations of buccal mucosa during function

In about 80% edentulous cases, food debris will not be trapped on the polished surface, regardless of how much concavity is given to allow the buccal mucosa to rest on. This way the buccal mucosa that rests on the denture will continue to the posterior direction, and it helps the posterior border seal with the buccal mucosa and the sidewall of the tongue resting on the retromolar pad more freely.

But some patients may complain that this kind of concavity will trap food debris on the buccal polished surface. Tooth loss will decrease not only the occlusal force, but also decrease the muscle force as well as the mucosal adaptation potential that is going to close in the associated gap. Aged edentulous patients with highly progressive bone resorption will hardly accept our instruction of muscle training, and so we give some convexity on the buccal polished surface in order to avoid the food debris trapping.

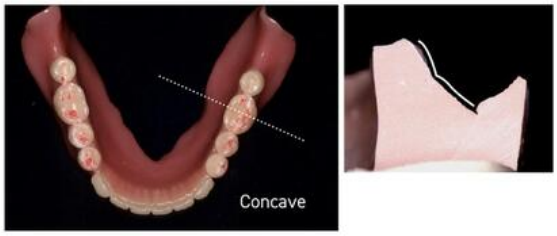
Clinical case (Figs 7-2,3)

A rather strong concavity is given to the posterior bucco-lingual polished surface of a denture that was made with us here about 8 years ago. After several regular checkups were done with us, no further appointments were made from 3 years ago. At the time of visit, a complaint was made that food debris was likely trapped on the lower posterior buccal polished surface. After the long absence we noticed some frailty of the patient. We diagnosed weakened oral strength and fewer cheek movements, and we decided to alter the polished surface morphology to a convex form to accommodate weakened function.



Fig 7-2 The complete denture used from 64 to 72 years of age: Concave buccal polished surface.

Concave



Convex

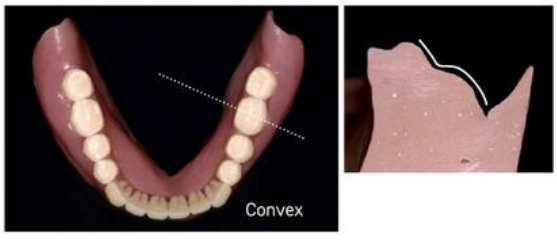


Fig 7-3a~d New denture: Convex form of the buccal polished surface.

[2] Information transfer to dental technician : Buccal polished surface morphology

Decision on whether a concavity or convexity regarding consideration of the buccal polished surface morphology should be given to a technician at the time of final impression taking. For example, check the mass of impression remaining on the custom tray after used to make a stone cast, or take a simple digital photo with written information of this and send to the technician. Impression of highly active buccal mucosa and that of low activity are shown here. They will help you to compare with your own clinical case (**Fig 7-4**).

a. Highly active buccal mucosa (provision of concavity)
b. Less active buccal mucosa (provision of convexity)

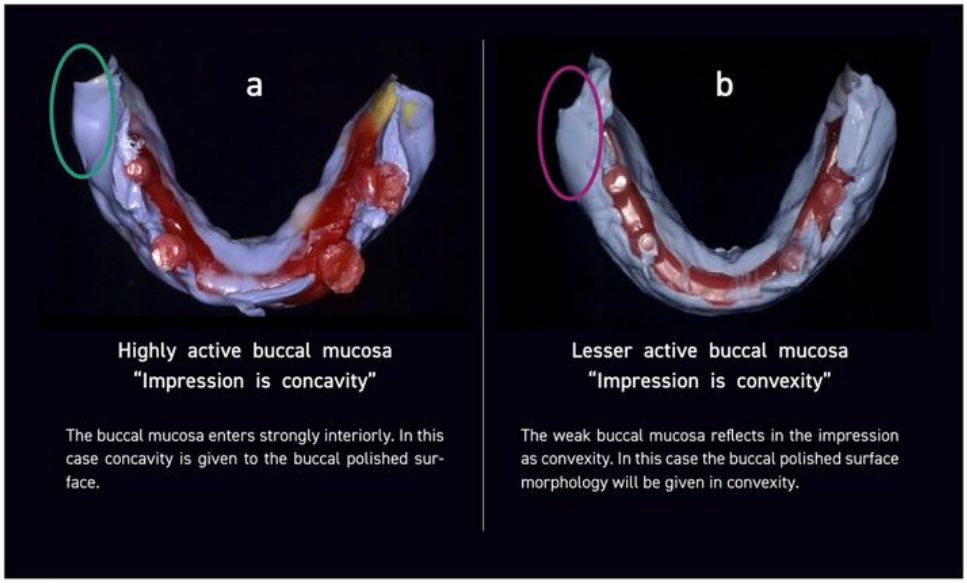


Fig 7-4

2 The basic contours of polished surfaces derived from function

This section describes the basic contours of the polished surfaces[40], as well as the shaping of the polished surfaces based on the ridge condition as observed on the casts when the technician has limited information. The method of reproducing the form of the polished surfaces from the final impressions with plaster cores will also be introduced.

Functionally shaped polished surfaces facilitate not only chewing, phonetics, esthetics and tongue sensation, but sealing of the denture borders. They are indispensable for achieving mandibular denture suction. The following two areas of polished surfaces are discussed.

(1) The polished surfaces that affect swallowing, phonetics and tongue posture (Figs 7-5,6).

(2) The labio buccal polished surfaces that vary depending on the level of cheek activity (Fig 7-7).

(1) The polished surfaces that affect swallowing, phonetics and tongue posture

The tongue is in light contact with the maxillary and mandibular anterior teeth at rest. The most important reason for creating an adequate space for the tongue is to keep the tongue in natural position. If tongue space is inadvertently reduced, tongue movement will be disturbed and the tongue will recede. This will result in a gap between the tongue and the anterior lingual denture border, and loss of border seal. The mandibular denture will lift easily with only a small mouth opening. It is possible to increase denture stability further by making the tongue-root areas of the denture base concave.



Fig 7-5a,b Waxing of the tongue-root areas (into concave shape) will facilitate achievement of suction by compensatory closure.



Fig 7-6a,b Waxing to provide a wide space for the tongue.

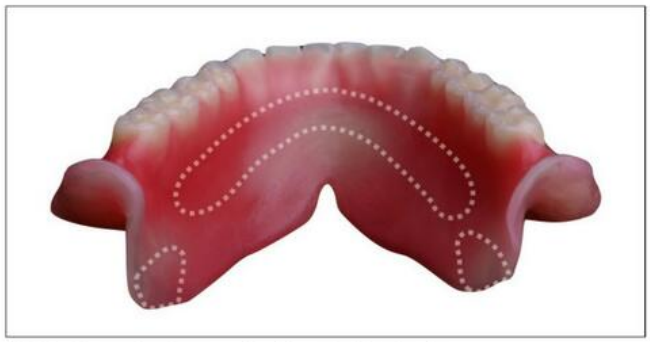


Fig 7-6c The completed wax-up of the lingual polished surfaces.

(2) The labio buccal polished surfaces that vary depending on the level of cheek activity.



Fig 7-7 The completed wax-up of the labiobuccal polished surfaces.

3 Wax denture try-in

[1] Analog wax denture try-in

Esthetic preferences vary from one patient to another. Some patients prefer average-colored well aligned teeth. Some patients like more individualised tooth colors. The number of patients who want white teeth has been increasing recently due to the popularity of tooth whitening. It is important to confirm the patient's preferences regarding the type of denture teeth and their alignment, and realise them during wax try-in (**Fig 7-8**).

In wax-denture try-in, we can check for esthetics, phonation, and denture stabilisation.

If the patient is not satisfied with wax dentures, tooth position can be easily changed then and there to enhance denture aesthetics with patient's preferences and increase the denture stability to reduce the lateral vectors in lateral and protrusive jaw movements.

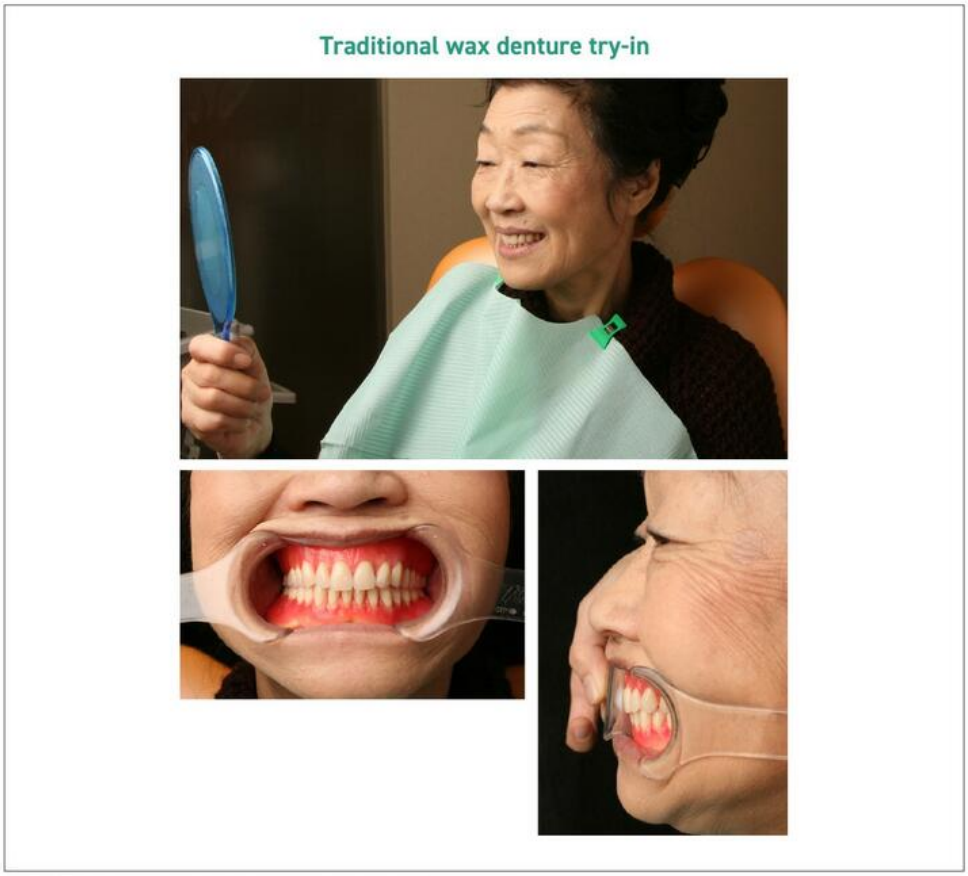


Fig 7-8 Tooth position can be easily changed with patient's preferences.

[2] Digital printed and milled denture try-in

ProArt CAD Try-In is a milky white PMMA disc (**Figs 7-9~11**). These mono block dentures allow us to carry out functional try-in because it has the ability of giving patients the chance to take these dentures home to try them out and also no issue of teeth falling off of the wax base plates (**Fig 7-11**).

Besides, we can check for the denture stability when the patient actually chews food and phonation that is related to the denture base thickness or thinness in addition to the check items for traditional wax denture try-in.



Fig 7-9

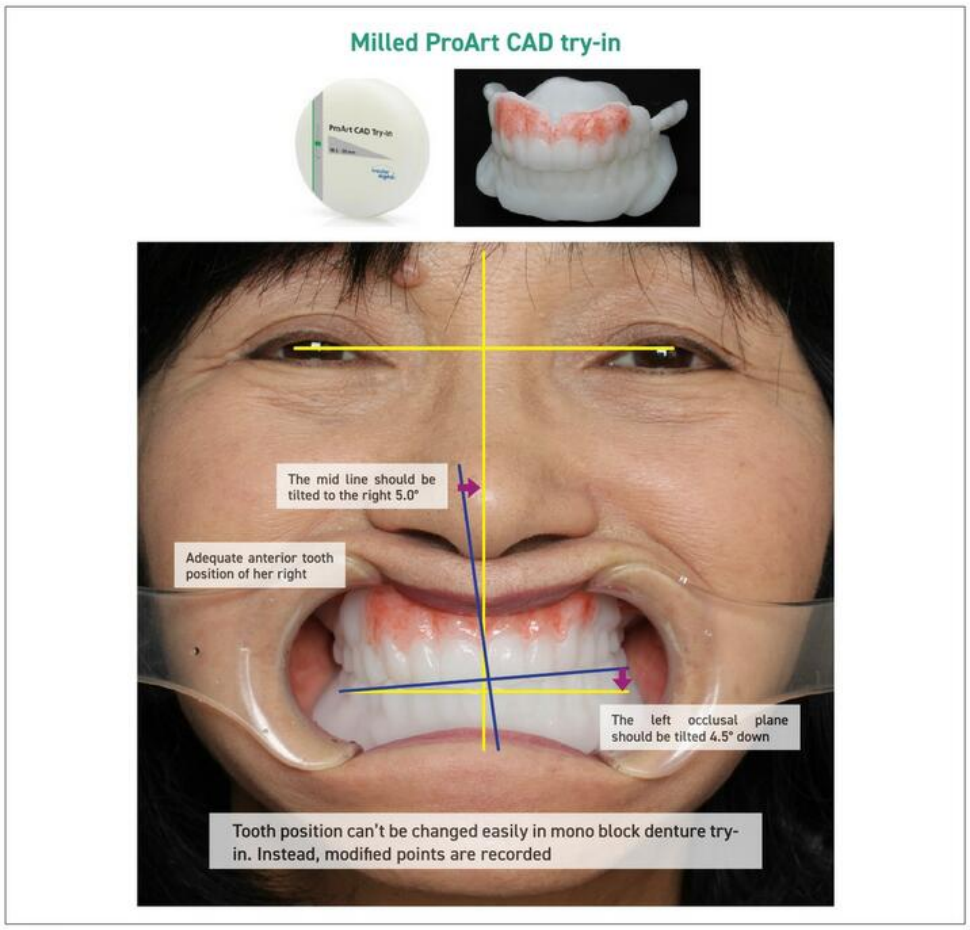


Fig 7-10

"Benefit of digital denture try-in"

It is available for the return when a patient wants a family to look at this denture at a home with putting it on.



Fig 7-11a Part painted red : Move the right first premolar and second premolar closer to the crest in order to make give sufficient over-bite.



Fig 7-11b Check the midline and horizontal line and facial configuration with the patient's preferences.

[3] Need for balancing contact (analog & digital)

Not only analog wax denture, but also digital printed or milled dentures allow us to check if the balanced occlusion functions properly without teeth popping off the try in.

When the posterior denture teeth contact only on one side, the other side of the mandibular denture lifts. This results in loss of retention. Likewise, when only the anterior denture teeth contact, the maxillary denture is pushed up and dislodged by the mandibular denture. When the patient bites hard into a thin piece of food or chew hard a bolus of food crushed by the posterior teeth, the denture on the working side will impinge on the alveolar ridge, producing a strong pain and preventing continued chewing.

To minimise the tipping or toppling of the dentures during chewing, the teeth on the balancing side must be brought into contact to redirect the toppling force into vertical compressive force (balancing contacts are required) (**Figs 7-13,14**). This occlusal scheme was developed from the assumption that the dentures move on the alveolar ridges, to minimise denture movement during chewing. It is called a bilateral balanced occlusion. The models in the figures were given a form of bilateral balanced occlusion with only the lingual cusps of the maxillary posterior teeth contacting the mandibular posterior teeth (lingualized occlusion).

Functional try-in of monoblock dentures

1) **Importance of balancing contacts in chewing for edentulous patients, a completely different situation from dentate individuals.**

2) **Balancing contacts are necessary because the dentures need to move on the alveolar ridges in order to function.**

Analog





Fig 7-13a,b When biting on a cotton roll on the right and left sides, the maxillary and mandibular denture teeth come into contact to stop the dentures from tipping.

Digital

Three dimension revision

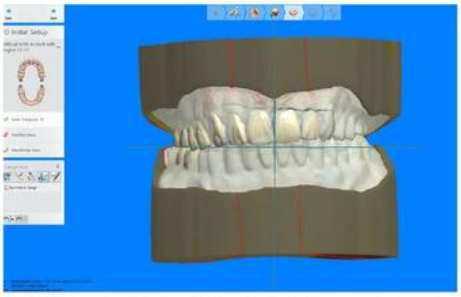


Fig 7-14a Inclined teeth of Monoblock, before the denture try-in.

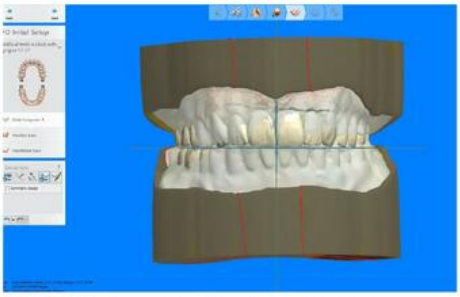


Fig 7-14b Final correction after alignment tooth set up.

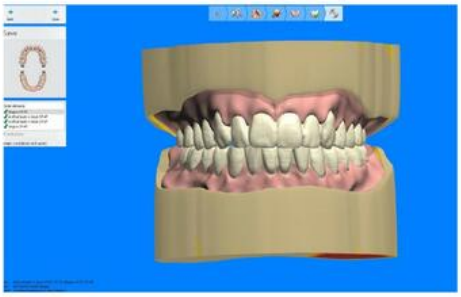


Fig 7-14c Final denture design.

4 Choice of resin and curing machine (Analog)

Dental stone is chosen according to the amount of shrinkage of the resin.

IvoBase System (Ivoclar Vivadent) is the auto curing system, which does not require a large scale equipment(**Fig 7-15**). Ideal resin polymerisation is controlled strictly by pre programmed temperature settings and hydraulic pressure (**Figs 7-16~18**).








Fig 7-15a~f Heavy body type (regular) of Virtual is used for the secondary investment. The stone does not adhere to cervical area at the time of deflasking. Mixed stone of Type IV stone and type II stone (2(IV):8(II)) is used for the third investment. Bonding strength is improved by applying resin primer 30 seconds before injection.



Fig 7-16a~c The optional Residual Monomer Reduction [RMR] function is not used, when the IvoBase Hybrid Pink is selected. Instead, the polymerization unit is pressurized with Polymax 5 (Dreve, Rinkai) under 6 atmospheric pressure at 55oC for 30minutes and then the denture is deflasked after gradual cooling for 12 hours.



Fig 7-17a,b Polymerization using the resin polymerizing apparatus is IvoBase injector. IvoBase Hybrid Pink is used for the denture base resin.



Fig 7-18a,b The sprue for the maxilla should be placed outside as shown in the photo for better fit instead of being placed on the alveolar ridge crest or inside.

5 Necessity of split cast (Analog)

A Split cast needs to be provided to the base of the model for remounting procedure after processing, as there may be a slight position change of the denture teeth due to shrinkage of resin after polymerisation is inevitable. A remounting procedure is required for occlusal adjustment to minimise clinical time allowing more time for a more detailed adjustment for each patient.

6 Occlusal adjustment after remounting (Analog)

1) maximum intercuspation
2) lateral movement (bilateral balanced occlusion)
3) protrusive movement

It is important to perform occlusal adjustment on the articulator beforehand to reduce the amount of time of required for intra-oral occlusal adjustment (**Fig 7-19**).

Occlusal adjustment after remounting



1) maximum intercuspation
2) lateral movement (bilateral balanced occlusion)
3) protrusive movement





Red marks : Contacts in maximum intercuspation after adjustment (lingualized occlusion).

Green marks: Contacts during lateral excursion after adjustment (bilateral balanced occlusion, lingualized occlusion).

Fig 7-19